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Hazards Identification, Risk Assessment and Control (HIRAC)

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Session Objective:

At the end of the session, the participants will be able to **perform** risks/hazards assessment, evaluation and recommend appropriate control measures.

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Risk Assessment process (HIRAC)

HIRAC or the Risk Assessment Process is the process of **identifying** workplace hazards, **evaluating risks** to workers' safety and health and **control** the relevant hazards.

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What is a risk assessment?

Risk assessment is a term used to describe the overall process or method where you:

- Identify hazards and risk factors that have the potential to cause harm (hazard identification).
- Analyze and evaluate the risk associated with that hazard (risk analysis, and risk evaluation).
- Determine appropriate ways to eliminate the hazard, or control the risk when the hazard cannot be eliminated (risk control).

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Why is risk assessment important?

Risk assessments are very important as they form an integral part of an occupational health and safety management plan. They help to:

- Create awareness of hazards and risk.
- Identify who may be at risk (e.g., employees, cleaners, visitors, contractors, the public, etc.).
- Determine whether a control program is required for a particular hazard.
- Determine if existing control measures are adequate or if more should be done.
- Prevent injuries or illnesses, especially when done at the design or planning stage.
- Prioritize hazards and control measures.
- Meet legal requirements where applicable.

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What is the goal of risk assessment?

The goal is to try to answer the following questions:

a. What can happen and under what circumstances?
b. What are the possible consequences?
c. How likely [and severe] are the possible consequences to occur?
d. Is the risk controlled effectively, or is further action required?

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When should a risk assessment be done?

There may be many reasons a risk assessment is needed, including:

- Before new processes or activities are introduced.
- Before changes are introduced to existing processes or activities, including when products, machinery, tools, equipment change or new information concerning harm becomes available.
- When hazards are identified.

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Risk assessment steps

European Agency of Safety and Health at Work	Health and Safety Executive
1. Identifying hazards and those at risk	1. Identify the hazards
2. Evaluating, prioritizing risks	2. Decide who might be harmed and how
3. Deciding on preventive action	3. Evaluate the risks and decide on precautions
4. Taking action	4. Record your significant findings
5. Monitoring and reviewing	5. Review your assessment and update if necessary

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Hazard identification

Process of finding and identifying:

- **hazardous agents** (situations, products etc.) that could contribute to provoking an occupational accident or/and disease
- the groups of **workers potentially exposed** to these hazards.

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Hazards, from where?

- A risk assessment must be carried out before young people start work, covering hazards and risk factors related to:

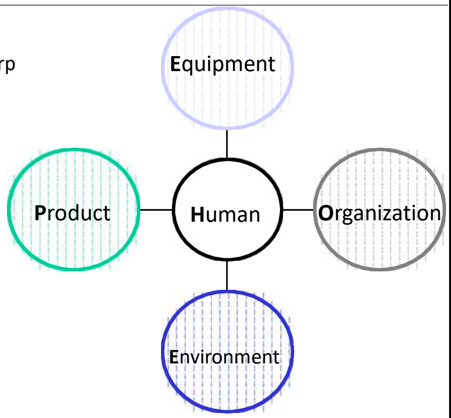
· **Equipment:** workplace lay-out, machines, hand tools, software and hardware, tables or chairs

· **Product:** dangerous substances, heavy loads and sharp or warm objects

· **Environment:** light, noise, climate, vibrations, air quality or dust

· **Organisation**: tasks, working hours, breaks, shift systems, training, communication, team work, contact with visitors, social support or autonomy.

· **Human:** lack of physical or mental capacity, lack of knowledge or skills, lack of right attitude or behavior



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Categories of Hazards

- **Safety Hazards** – something that has potential to cause injury
  - Poor housekeeping
  - Fire
  - Use of machine
  - Material handling
  - Electricity
- **Health Hazards** – Any agent or activity posing potential hazard to health
  - Chemical (vapors, mists fumes, gases, dusts)
  - Physical (noise, vibration, illumination, extreme temperature, extreme pressure, radiation)
  - Biological (bacteria, viruses, molds, fungi, protozoa; and insects, parasites, plants, animals
  - Ergonomics (improperly designed tools or work areas, improper lifting or reaching, poor visual conditions, repeated motion in awkward position

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How are the hazards identified?

- Walk through/ocular inspection
- Review of Processes involved
- Knowing the raw materials used, products and by-products
- Gathering workers' complaints
- Safety data sheet

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How are the hazards identified?

To be sure that all hazards are found:

- Include non-routine activities such as maintenance, repair, or cleaning.
- Look at accident / incident / near-miss records.
- Include people who work off site either at home, on other job sites, drivers, teleworkers, with clients, etc.
- Look at the way the work is organized or done (include experience of people doing the work, systems being used, etc).
- Look at foreseeable unusual conditions (for example: possible impact on hazard control procedures that may be unavailable in an emergency situation, power outage, etc.).

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How are the hazards identified? (cont.)

To be sure that all hazards are found:

- Determine whether a product, machine or equipment can be intentionally or unintentionally changed (e.g., a safety guard that could be removed).
- Examine risks to visitors or the public.
- Consider the groups of people that may have a different level of risk such as young or inexperienced workers, persons with disabilities, or new or expectant mothers.

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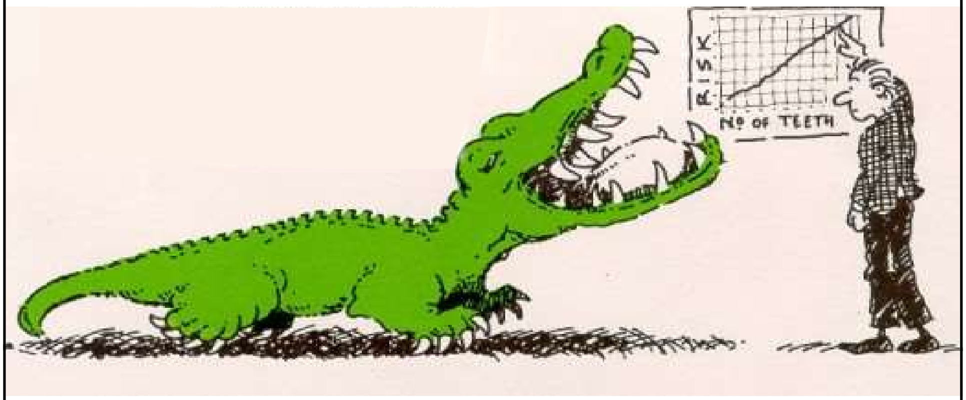
Evaluation of risk



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How do you know if the hazard will cause harm (poses a risk)?

Each hazard should be studied to determine its' level of risk. To research the hazard, you can look at:

- Product information / manufacturer documentation.
- Past experience (knowledge from workers, etc.).
- Legislated requirements and/or applicable standards.
- Industry codes of practice / best practices.
- Health and safety material about the hazard such as safety data sheets (SDSs), research studies, or other manufacturer information.
- Information from reputable organizations.

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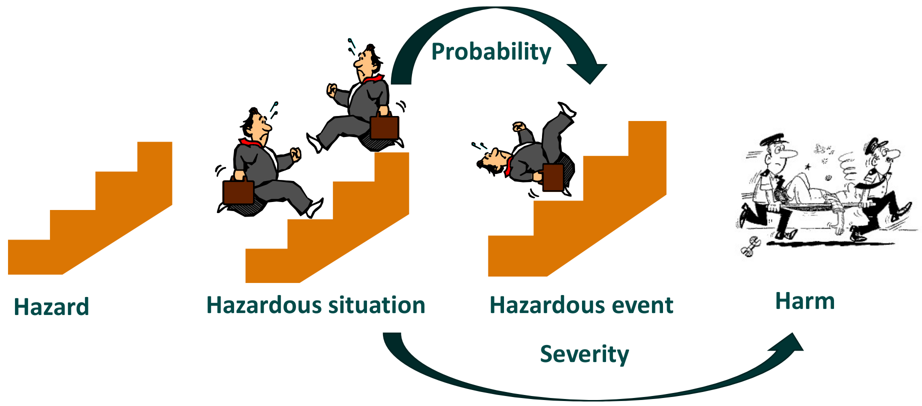
Evaluation of risk

(**Risk**) = (**Probability**) of event × (**Severity**) of harm



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How do you know if the hazard will cause harm (poses a risk)? Cont.

Each hazard should be studied to determine its' level of risk. To research the hazard, you can look at: (cont.)

- Results of testing (atmospheric or air sampling of workplace, biological swabs, etc.).
- The expertise of an occupational health and safety professional.
- Information about previous injuries, illnesses, near misses, incident reports, etc.
- Observation of the process or task.

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Medical Surveillance

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Purpose of Medical Surveillance

PRIMARY

- prevention of illness.
  - Aims of Occupational Health (WHO, ILO)

SECONDARY

- early detection of work-related health problems and determining its cause.

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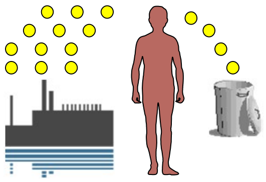
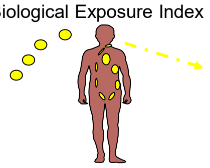
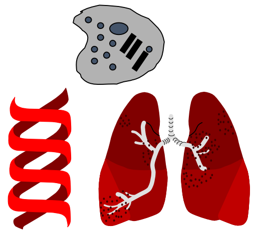
Monitoring Strategies

Ambient monitoring	Biological monitoring	Health surveillance
• measurement and assessment of agents at the workplace • evaluates ambient exposure compared to reference • Threshold Limit Value	• measurement and assessment of agents or their metabolites either in tissues, secreta, excreta, expired air, or any combination • evaluates exposure compared to reference • Biological Exposure Index	• periodic medico-physiologic examinations of exposed workers • aims to protect health and prevent occupationally related disease
	 Absorption, metabolism, distribution, elimination	

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Monitoring Strategies

Monitoring Activity	Workplace-Related Events	Health Effects
Environmental monitoring	Exposure at the workplace - Chemicals - Physical agents	None
Biological monitoring and Medical surveillance	Biologically significant exposure - Chemicals absorbed - Early (reversible) changes	Early
	Clinical diagnosis - measurable health effects	
Treatment and surveillance (morbidity, mortality)	End effects - diseases - unfavorable events (spontaneous abortion)	Late

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Why do we conduct Medical Surveillance?

- Worker populations are not homogenous
  - Certain members will be particularly vulnerable or at greater risk than others. *(individual susceptibility)*

- Hazard level within safe or acceptable limits do not guarantee that workers will not be affected.

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Remember to include factors that contribute to the level of risk such as:

- The work environment (layout, condition, etc.).
- The systems of work being used.
- The range of foreseeable conditions.
- The way the source may cause harm (e.g., inhalation, ingestion, etc.).
- How often and how much a person will be exposed.
- The interaction, capability, skill, experience of workers who do the work.

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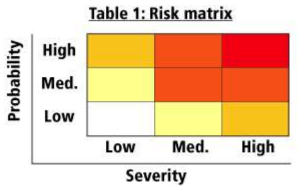
How are risks ranked or prioritized?



Probability ratings in this example represent:

- **High:** likely to be experienced once or twice a year by an individual
- **Medium:** may be experienced once every five years by an individual
- **Low:** may occur once during a working lifetime

Severity ratings in this example represent:

- **High:** major fracture, poisoning, significant loss of blood, serious head injury, or fatal disease
- **Medium:** sprain, strain, localized burn, dermatitis, asthma, injury requiring days off work
- **Low:** an injury that requires first aid only; short-term pain, irritation, or dizziness

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How are risks ranked or prioritized? (cont.)

Table 2: Risk Ratings

Description	Colour Code
Immediately Dangerous	
High Risk	
Medium Risk	
Low Risk	
Very Low Risk	

- These risk ratings correspond to recommended actions such as:
  - **Immediately dangerous:** stop the process and implement controls
  - **High risk:** investigate the process and implement controls immediately
  - **Medium risk:** keep the process going; however, a control plan must be developed and should be implemented as soon as possible
  - **Low risk:** keep the process going, but monitor regularly. A control plan should also be investigated
  - **Very low risk:** keep monitoring the process

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Example of Risk Assessment				
Task	Hazard	Risk	Priority (L/M/H)	Control
Painting a room	Stepping on a 1 meter step tool to reach higher areas	Falling from 1 meter height Severity: cause a short term strain or sprain. A severe sprain may require a few days off work. Probability: occur once in a lifetime as painting is an uncommon activity in this organization		

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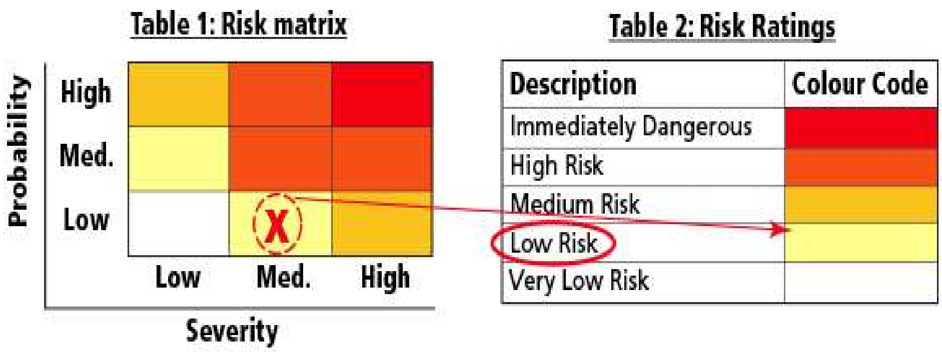
How are risks ranked or prioritized? (cont.)



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Example of Risk Assessment				
Task	**Hazard**	**Risk**	**Priority (L/M/H)**	**Control**
Painting a room	Stepping on a 1 meter step tool to reach higher areas	Falling from 1 meter height Severity: cause a short term strain or sprain. A severe sprain may require a few days off work. Probability: occur once in a lifetime as painting is an uncommon activity in this organization	L	Use of stool with a large top to maintain stability.

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CLASSIFICATION OF RISK LEVELS

Low risk establishment – refers to a workplace where there is low level of danger or exposure to safety and health hazards and not likely or with low probability to result in accident, harm, injury, or illness.

Medium risk establishment – refers to a workplace where there is moderate exposure to safety and health hazards and with probability of an accident, injury or illness, if no preventive or control measures are in place.

High risk establishment – refers to a workplace where there is high level of exposure to safety and health hazards, and probability of a major accident resulting to disability, death or major illness is likely to occur if no preventive or control measures are in place.

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The following are workplaces commonly associated with potentially **high-risk activities**:

1. Chemical works and chemical production plants;
2. Construction;
3. Deep sea fishing;
4. Explosives and pyrotechnics factories;
5. Firefighting;
6. Healthcare facilities;
7. Installation of communication accessories, towers and cables;
8. LPG filling, refilling, storage and distribution;
9. Mining;
10. Petrochemical works and refineries;
11. Power generation, transmission and distribution in the energy sector;
12. Storage and distribution center for toxic or hazardous chemicals;
13. Storage of fertilizers in high volume;
14. Transportation;
15. Water supply, sewerage, waste management, remediation activities;
16. Works in which chlorine is used in bulk; and
17. Activities closely similar to those enumerated above and other activities as determined by DOLE in accordance with existing issuances on the classification of establishments.

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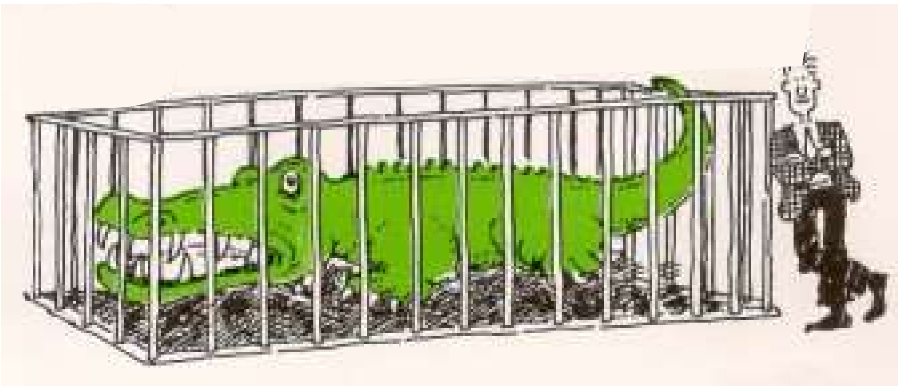
Control of Hazards



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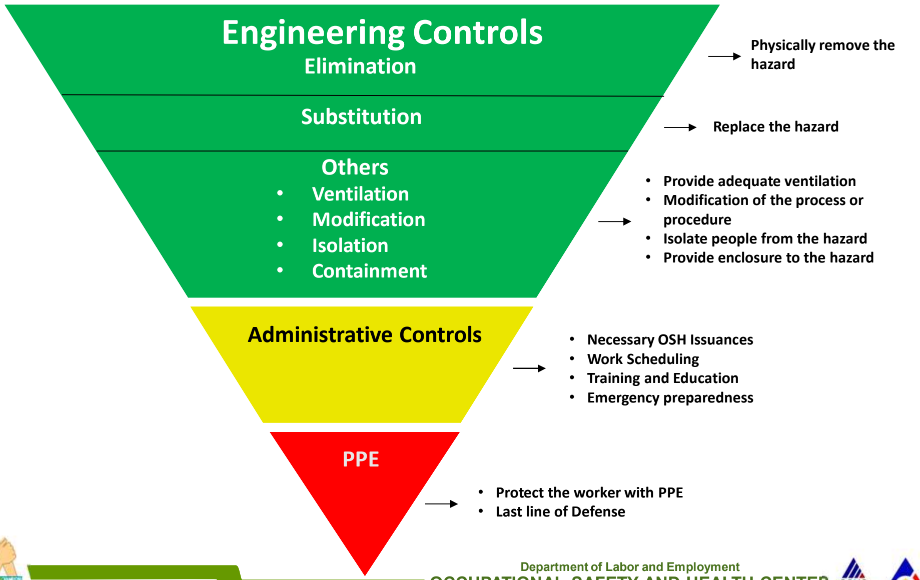
Hierarchy of Controls



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Workshop

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Workshop on the Use of Risk Assessment Tool

Watch video: METAL WORKS

1. Hazard Identification				2. Risk Analysis/Evaluation				3. Risk Control	
No	Work Activity	Hazard	Which can cause/effect	Existing Risk Control (if any)	Proba-bility	Severity	Risk	Proposed Control Measure	Due Date/ Status
1	Cutting	a. Safety Hazards 1. 2. b. Health Hazards 1. 2.							
2	Polishing	a. Safety Hazards b. Health Hazards							
3	Welding	a. Safety Hazards b. Health Hazards							
4	Painting	a. Safety Hazards b. Health Hazards							

Department of Occupational Safety and Health Ministry of Human Resources Malaysia *www.dosh.gov.my/images/dmdocuments/glx/ve_gl_hirarc.pdf*

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Review

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Review questions

1. What are the three major steps in HIRAC?

2. Give the formula in calculating risk?

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CLEAR POINTS

- When conducting hazard identification, it may help to work as a team and include both people familiar with the work area, as well as people who are not - this way you have both the experienced and fresh eye to conduct the inspection.

- There is no one simple or single way to determine the level of risk. Nor will a single technique apply in all situations. The organization has to determine which technique will work best for each situation.

- Ranking or prioritizing hazards is one way to help determine which risk is the most serious and thus which to control first.

- Any violation of the OSH Standards and other laws must be considered as immediately dangerous or totally unacceptable.

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